1 Methods

1.1 Fine-scale H&E–MALDI-MSI alignment

We perform fine-scale alignment of MALDI-MSI measurements to H&E histology after an initial fiducial-based registration. Let $\mathcal{N} = \{1, \ldots, n\}$ denote the MSI spots and

$$\Omega_{\mathrm{MSI}} = \{u_1, \ldots, u_n\} \subset \mathbb{R}^2$$

their positions in the original MSI coordinate system. The preprocessed MSI data are collected in the spot-by-feature matrix $X \in \mathbb{R}^{n \times p}$. Low-quality and low-variance features are removed, and the retained features are centred and scaled to mean zero and variance one across the complete slide. This standardisation is kept fixed throughout the alignment procedure.

An initial transformation

$$T_0 : \Omega_{\mathrm{MSI}} \to \mathbb{R}^2$$

maps MSI coordinates into the H&E coordinate system. It is obtained from fiducial markers and provides the starting geometry for subsequent fine alignment. Rather than attempting to register the complete H&E image directly to the high-dimensional MSI measurements, we derive a scalar anatomical pattern from H&E and identify a corresponding linear MSI metafeature. Geometry is then optimised by maximising agreement between these two spatial patterns. This reduces the multimodal registration problem to the alignment of two informative scalar fields while retaining the molecular information contained across all MSI features.

For liver tissue, the H&E reference is a sinusoid mask S_{HE}. Sinusoids provide dense spatial structure at a scale somewhat larger than an individual MSI measurement and are therefore well suited to correcting residual errors of the fiducial-based registration. The construction of this anatomical mask is described in Supplementary Section 2.1.

Because H&E has substantially higher spatial resolution than MSI, the binary mask is not compared directly with individual MSI coordinates. Instead, it is smoothed at approximately the MSI sampling scale using a compactly supported Wendland kernel and locally normalised with respect to the valid tissue mask. This gives a continuous H&E field $f : \mathbb{R}^2 \to \mathbb{R}$, described in Supplementary Section 2.2. For any transformation T, its projection to the MSI measurements is

$$s_{T,i} = f\big(T(u_i)\big), \qquad i \in \mathcal{N}.$$

This smoothing-and-projection step represents the H&E signal at the effective spatial scale of MSI rather than treating sub-MSI image structure as directly resolved.

The initial target s_{T_0} is used to identify a linear MSI metafeature. We estimate a ridge-regularised direction $w \in \mathbb{R}^p$ whose score Xw is associated with the anatomical target. With a one-dimensional target, this is equivalent to a regularised canonical association problem and can be expressed directly as ridge regression [1]. The regression formulation and normalisation of the resulting metafeature direction are given in Supplementary Section 2.3.

Importantly, this metafeature is not initially learned from the complete slide. The fiducial transformation T_0 is most reliable close to the fiducial markers, whereas larger registration errors may occur farther away. We therefore construct peri-fiducial training neighbourhoods (Supplementary Section 2.4) and determine their extent by repeated spatially balanced fiducial-level validation (Supplementary Section 2.5). This asks whether an MSI metafeature learned around one spatially distributed subset of fiducials predicts the H&E target around independent fiducials. After selection of the peri-fiducial radius, the metafeature is re-estimated from all selected peri-fiducial spots and evaluated over the complete slide,

$$q^{(0)} = X\hat{w}^{(0)}.$$

The molecular pattern $q^{(0)}$ is then held fixed during geometric optimisation. Thus, the molecular representation is learned from regions where the initial geometry is already reliable and is not allowed to adapt subsequently to registration errors elsewhere on the slide.

Residual broad deformation is estimated by a smooth thin-plate spline (TPS),

$$T_1(u) = T_0(u) + \widehat{\Delta}^{\mathrm{TPS}}(u).$$

The TPS maximises the correlation between the fixed MSI pattern $q^{(0)}$ and the projected H&E target while penalising bending of the displacement field. We use a deliberately coarse low-rank TPS basis and calibrate its regularisation geometrically by requiring residual motion in the peri-fiducial regions to remain small. The complete parameterisation and calibration procedure are described in Supplementary Section 2.6. This TPS step provides the principal fine-scale alignment and is sufficient for most of the tissue.

A globally smooth transformation can, however, be inappropriate at physical tissue tears, where neighbouring tissue regions may have moved relative to one another. We therefore inspect the TPS displacement field for unusually large local strain and combine this geometric signal with a permissive H&E brightness criterion to identify tear-associated regions. Spots outside these regions form the trusted set $\mathcal{G}$, whereas tear-associated spots form the movable set $\mathcal{U}$, with

$$\mathcal{N} = \mathcal{G} \,\dot{\cup}\, \mathcal{U}.$$

The tear-detection procedure, including construction of the auxiliary tear-support set and tear skeleton, is described in Supplementary Section 2.7.

Within the tear-associated regions, we allow an additional local movement of the non-trusted spots while keeping all trusted spots fixed. The displacement field is estimated by maximising the correlation between the fixed MSI metafeature $q^{(0)}$ and the projected H&E target, subject to a weighted graph-Laplacian penalty on neighbouring displacements. The regularisation is strongest close to trusted tissue and is progressively relaxed towards the tear according to the relative Euclidean distances to the trusted region and the tear skeleton. Graph connections across the tear-support region are removed, allowing the two sides of a tear to move independently while retaining a smooth displacement field within intact tissue.

The local regularisation strength is calibrated using trusted tissue surrounding the detected tear regions: the tear-associated regions are expanded into neighbouring trusted tissue to define a validation area, and the weakest regularisation is chosen for which the same local refinement produces only sub-pixel motion in this already trusted region. Tear-support spots and disconnected fragments for which no locally anchored refinement can be inferred remain unrefined. The complete optimisation, mobility weighting, calibration and quality-control measures are described in Supplementary Section 2.8.

1.2 Identification and spatial validation of a hepatocyte nucleus-associated MSI metafeature

We next ask whether the sinusoid-guided alignment reaches a spatial precision sufficient to detect molecular variation associated with individual hepatocyte nuclei. This provides both a stringent assessment of the achieved registration resolution and a biological application of the aligned data.

A nucleus mask is poorly suited as the initial fine-registration target. The diameter of a hepatocyte nucleus is of the same order as the MSI sampling scale, such that a residual registration error of approximately one MSI pixel can already remove much of the spatial correspondence between nuclear occupancy and the molecular measurement. In contrast, the sinusoidal network contains structures extending over several MSI measurements and can therefore be recognised while residual registration uncertainty is still larger. We consequently use the sinusoid-derived signal to establish the fine-scale alignment before searching for molecular variation at the nuclear scale.

To make this analysis conservative with respect to registration, we use only the trusted spots $\mathcal{G}$, which require no tear-specific correction and remain at their TPS-aligned positions

$$p_i = T_1(u_i), \qquad i \in \mathcal{G}.$$

The nucleus-associated analysis is therefore independent of the additional tear-aware refinement of the movable region.

Let M_{hep} and M_{nuc} denote H&E-derived masks of hepatocyte tissue and nuclei. Using the same locally normalised Wendland projection employed for registration, we calculate for each trusted MSI spot a hepatocyte fraction h_i and the nuclear occupancy y_i within its hepatocyte-containing footprint. We retain only hepatocyte-rich measurements,

$$\mathcal{I} = \{i \in \mathcal{G} : h_i \geq 0.9\}.$$

Thus, variation caused by vessels, immune-cell aggregates and other non-hepatocyte compartments is excluded before searching for a nuclear molecular signal.

The aim is to identify MSI features that are locally enriched or depleted at hepatocyte nuclei relative to the surrounding hepatocyte background. Broad spatial MSI variation, in particular periportal–pericentral zonation, could otherwise induce an association with nuclear occupancy that does not reflect nuclear localisation. We therefore subtract a local linear spatial background from each MSI feature, using

$$\sigma_{\text{bg}} = 4d_{\text{MSI}},$$

where d_{MSI} is the typical distance between neighbouring MSI measurements. The nuclear occupancy y_i itself is not high-pass filtered: it is obtained from a segmented nuclear mask and is the biological quantity that the local MSI contrast is intended to predict.

A weighted ridge regression then identifies a linear combination of the background-corrected MSI features associated with continuous nuclear occupancy. The weights balance nuclear and non-nuclear hepatocyte material without converting the continuous target into a binary class. Background-corrected MSI features are centred and scaled using weighted means and variances estimated from the respective training data. After fitting, the coefficients are transformed back to the original standardised MSI feature scale, yielding

$$q_{\text{nuc}} = X\hat{w}_{\text{nuc}}.$$

Spatial background correction and weighted standardisation are therefore used only to identify the metafeature direction; the reported nucleus-associated metafeature is an ordinary linear combination of the original MSI measurements. Details of target construction, spatial background correction, weighting, standardisation and metafeature estimation are given in Supplementary Section 2.9.

We next assess directly whether this metafeature can localise hepatocyte nuclei in spatially independent tissue. The metafeature direction is learned from spatially separated training regions and then applied to background-corrected MSI measurements in the corresponding held-out regions. All feature means and standard deviations used for standardisation are estimated exclusively from the training region and applied unchanged to the held-out MSI measurements.

After only tight spatial smoothing, local maxima of the held-out background-corrected nucleus-associated MSI score are used as candidate nucleus positions. A peak-height threshold is learned in the corresponding training region by maximising the object-level F1 score and is fixed before evaluation of the held-out tissue. In addition, precision–recall curves over the complete range of peak-height thresholds are reported to characterise the dependence of nucleus detection on the selected operating threshold.

Predicted MSI nucleus positions are matched one-to-one to H&E-derived hepatocyte nucleus centres in the held-out regions, with a maximum matching distance of one MSI spacing. We report sensitivity, positive predictive value and F1 score. Specificity is not reported, because point detection provides no natural definition of the number of true-negative spatial positions; any specificity would therefore depend on an arbitrary discretisation of the image rather than on the localisation problem itself.

For correctly matched nuclei, the Euclidean distance between the MSI-derived peak and the corresponding H&E nucleus centre provides a direct measure of localisation accuracy. We report the distribution of these distances, including its median and 95th percentile, the fractions of matched nuclei localised within half and one MSI spacing, and the mean two-dimensional localisation error as a measure of systematic residual displacement.

Importantly, failure to detect an individual nucleus is not necessarily a registration error: a nucleus-associated metabolite need not exhibit the same degree of enrichment at every hepatocyte nucleus. Sensitivity therefore reflects both registration quality and the biological consistency of the molecular signal. In contrast, the localisation error among correctly detected nuclei provides the more direct measure of the spatial precision of the H&E–MSI correspondence.

As a complementary validation that does not depend on discrete peak calling, we systematically translate the H&E nuclear-occupancy pattern relative to the held-out background-corrected MSI score and recalculate their correlation. The resulting two-dimensional shift-response surface should attain its maximum close to zero displacement if the inferred registration is accurate at approximately the nuclear scale. We report the complete response surface and the displacement at which its maximum occurs. The spatial splitting, nucleus calling, one-to-one matching, localisation-error analysis and shift-response analysis are described in Supplementary Section 2.10.

Together, object-level nucleus localisation and the shift-response surface assess whether the sinusoid-guided alignment preserves molecular–histological correspondence at approximately the nuclear scale. The resulting metafeature is interpreted as reflecting metabolites locally enriched or depleted at hepatocyte nuclei relative to the surrounding hepatocyte background. Because the MSI footprint is itself comparable in size to a nucleus, closely associated perinuclear and intranuclear signals cannot be distinguished by this analysis.

2 Supplementary Methods

2.1 Generation of the H&E-derived anatomical mask

Let

$$M : \Omega_{\mathrm{HE}} \to \{0, 1\}$$

denote the valid H&E tissue mask, with $M(z) = 1$ for pixels belonging to foreground tissue considered in the analysis.

In liver tissue, sinusoids provide an informative anatomical marker for fine-scale alignment of H&E and MSI. In two-dimensional sections they form bright, elongated structures with characteristic organisation on the scale of a few cells. The H&E image is converted to HSV colour coordinates, and a permissive empirical criterion based on brightness and hue is used to identify candidate sinusoid pixels. Exterior background is removed, as are pixels belonging to larger vessels identified by Cellpose, StarDist or an equivalent validated segmentation procedure. This yields the anatomical mask

$$S_{\mathrm{HE}} : \Omega_{\mathrm{HE}} \to [0, 1].$$

For the sinusoid application used here, S_{HE} is binary. The more general range $[0, 1]$ is retained in the notation because the same alignment framework can use a soft H&E-derived marker.

For tissues in which a sinusoid mask is inappropriate, the same procedure can be applied to another scalar anatomical signal, for example a vessel mask, nuclei-density map, image-gradient or edge response, blob-detector response, or a selected component from a PCA or non-negative matrix factorisation of H&E-derived image features. The essential requirement is not the biological identity of the marker but that it provide reproducible spatial contrast expected to have a corresponding molecular signature.

2.2 Wendland smoothing and projection of the H&E mask

A transformed MSI position will generally not coincide with an H&E pixel, and the native H&E mask contains spatial detail far below the MSI sampling scale. We therefore convert S_{HE} into a smooth field before evaluating it at transformed MSI positions.

For support radius $\rho > 0$, we use the compactly supported Wendland C^2 kernel

$$K_\rho(x) = \frac{1}{C_\rho}\left(1 - \frac{\|x\|_2}{\rho}\right)_+^4 \left(4\frac{\|x\|_2}{\rho} + 1\right),$$

where $(a)_+ = \max(a, 0)$, and C_ρ is chosen such that

$$\int_{\mathbb{R}^2} K_\rho(x)\,\mathrm{d}x = 1.$$

The support radius is tied to the MSI sampling geometry. Let E_4 denote the set of direct horizontal or vertical neighbour pairs on the original rectangular MSI grid. We set

$$\rho = \frac{\sqrt{2}}{2}\,\mathrm{median}_{(i,j)\in E_4} \left\|T_0(u_i) - T_0(u_j)\right\|_2.$$

For an approximately square MSI lattice, this is half the diagonal of a typical grid cell. It therefore provides continuous coverage between neighbouring measurements while keeping the kernel footprint close to the experimental sampling scale.

Both M and S_{HE} are extended by zero outside the H&E image on a common rectangular computational grid. Ordinary zero-padded convolution would attenuate the signal close to tissue boundaries. We therefore divide by the locally available tissue mass and define

$$f(z) = \frac{\big(S_{\mathrm{HE}} * K_\rho\big)(z)}{\big(M * K_\rho\big)(z)}$$

where the denominator is positive, and set $f(z) = 0$ otherwise. This is a locally normalised, Nadaraya–Watson-type kernel estimate. Bilinear interpolation of the resulting grid values extends f to $\mathbb{R}^2$.

For any transformation T, projection of the smoothed H&E mask onto the MSI measurements gives

$$s_T = \big(f\big(T(u_i)\big)\big)_{i\in\mathcal{N}}\,.$$

The same field f is retained throughout registration; only the locations at which it is evaluated change.

2.3 MSI metafeature learning on a specified set of spots

The initial transformation T_0 gives the anatomical target s_{T_0}. We seek a linear MSI score whose spatial pattern has a stable positive association with this target. Our intended criterion is a ridge-regularised canonical association between the MSI feature space and the one-dimensional anatomical signal [1]. With a single target variable, the optimisation can be expressed directly as ridge regression, which we use from this point onwards.

For an arbitrary index set $\mathcal{I} \subseteq \mathcal{N}$, let $X_{\mathcal{I}}$ and $s_{T_0,\mathcal{I}}$ denote the corresponding rows of X and entries of s_{T_0}. We estimate an unpenalised intercept and ridge coefficients by

$$\left(\hat\alpha_{\mathcal{I}}, \hat\beta_{\mathcal{I}}\right) = \arg\min_{\alpha,\beta} \left\{ \frac{1}{|\mathcal{I}|} \, \|s_{T_0,\mathcal{I}} - \alpha\mathbf{1} - X_{\mathcal{I}}\beta\|_2^2 + \lambda_{\mathrm{ridge}}\|\beta\|_2^2 \right\}, \tag{1}$$

with fixed

$$\lambda_{\mathrm{ridge}} = 0.05.$$

The intercept absorbs the mean of the anatomical target and any shift in feature means induced by restricting the globally standardised matrix X to $\mathcal{I}$. The metafeature direction is

$$\hat w_{\mathcal{I}} = \frac{\hat\beta_{\mathcal{I}}}{\|\hat\beta_{\mathcal{I}}\|_2}. \tag{2}$$

Its sign is chosen such that

$$\operatorname{cor}\left(X_{\mathcal{I}}\hat w_{\mathcal{I}}, s_{T_0,\mathcal{I}}\right) \geq 0.$$

The ridge term suppresses unstable directions formed from highly correlated or weakly supported MSI features. Because the columns of X have unit variance on the complete slide, the fixed value $\lambda_{\mathrm{ridge}} = 0.05$ is small relative to the marginal feature variance while still regularising nearly singular feature combinations. We keep this penalty fixed rather than optimising it against spatial agreement, thereby avoiding another opportunity for the molecular target to adapt to registration error.

2.4 Construction of peri-fiducial neighbourhoods

The initial transformation is expected to be most reliable close to the fiducial markers. We therefore determine how far metafeature learning can be extended away from them without losing reproducibility.

Let

$$h_1, \dots, h_F \in \Omega_{\mathrm{HE}}$$

denote the F fiducial coordinates in the H&E image. Each MSI spot is assigned to its nearest fiducial under the initial transformation,

$$\kappa_i = \arg \min_{1 \leq k \leq F} \| T_0(u_i) - h_k \|_2 \,,$$

and we define the corresponding distance

$$d_i^{\mathrm{fid}} = \| T_0(u_i) - h_{\kappa_i} \|_2 \,.$$

Ties are resolved deterministically. The ownership assignment κ_i is fixed for all candidate radii. Consequently, regions belonging to different fiducial folds remain disjoint even if geometric discs around the fiducials overlap.

For fiducial k, let

$$m_k = |\{ i \in \mathcal{N} : \kappa_i = k \}|$$

denote the size of its complete nearest-fiducial territory.

We generate $B = 10$ repeated bipartitions

$$A_b \,\dot{\cup}\, B_b = \{1, \dots, F\}, \qquad b = 1, \dots, B.$$

Each split is constructed to balance the number of assigned MSI spots while distributing both halves over the slide. The split construction is independent of the candidate radius, so the same fiducial split is used for all radii.

For each repetition, one fiducial is sampled uniformly and assigned to A_b, and a second distinct fiducial is sampled uniformly and assigned to B_b. Define the current territory sizes

$$m_{A_b} = \sum_{k \in A_b} m_k, \qquad m_{B_b} = \sum_{k \in B_b} m_k.$$

While unassigned fiducials remain, the next fiducial is added to the set with the smaller current territory size, with ties assigned to A_b. To favour spatial coverage, the fiducial added to a given set $C \in \{A_b, B_b\}$ is sampled according to a k-means++ rule [2]. If $\mathcal{K}_{\mathrm{rem}}$ is the set of unassigned fiducials, then fiducial $k \in \mathcal{K}_{\mathrm{rem}}$ is sampled with probability proportional to

$$w_C(k) = \min_{j \in C} \| h_k - h_j \|_2^2.$$

This favours addition of fiducials far from those already present in the same fold.

For any fiducial subset $A \subseteq \{1, \dots, F\}$ and radius r, define

$$\mathcal{I}_A(r) = \left\{ i \in \mathcal{N} : \kappa_i \in A,\ d_i^{\mathrm{fid}} \leq r \right\}, \qquad n_A(r) = |\mathcal{I}_A(r)|.$$

The corresponding target variance is

$$v_A(r) = \mathrm{Var} \left\{ s_{T_0,i} : i \in \mathcal{I}_A(r) \right\}.$$

For the complete slide, define

$$v_{\text{all}} = \operatorname{Var}\{s_{T_0,i} : i \in \mathcal{N}\}.$$

The lower end of the radius search is the smallest radius $r_{\min}$ for which both halves of every fiducial split satisfy

$$n_A(r) \geq \max(3p, 0.05 \cdot |\mathcal{N}|) \qquad \text{and} \qquad v_A(r) \geq \frac{1}{2} v_{\text{all}}.$$

The first condition ensures that a training sample has appreciably more spots than the number of MSI features and at least 5% of all spots, whereas the second excludes tiny regions in which the anatomical target has too little variation to identify a meaningful spatial association. These criteria only define the search range; they are not used as performance measures.

The radius required to cover every valid MSI spot is

$$r_{\max} = \max_{i \in \mathcal{N}} d_i^{\text{fid}}.$$

The candidate set contains $r_{\min}$, $r_{\max}$, and eight intermediate empirical quantiles of the distances d_i^{fid} lying between these bounds.

2.5 Selection of the peri-fiducial radius by split-set validation

For candidate radius r and split b, we fit $\hat{w}_{A_b}(r)$ using $\mathcal{I}_{A_b}(r)$ and evaluate the resulting score on the complementary fiducial territories,

$$C^{(b)}_{A \to B}(r) = \operatorname{cor}\left(X_{\mathcal{I}_{B_b}(r)} \hat{w}_{A_b}(r), s_{T_0, \mathcal{I}_{B_b}(r)} \right).$$

Reversing the roles of the two halves gives $C^{(b)}_{B \to A}(r)$. The two directional correlations are combined on the Fisher scale,

$$Z_b(r) = \frac{1}{2} \left\{ \operatorname{atanh}\left[C^{(b)}_{A \to B}(r) \right] + \operatorname{atanh}\left[C^{(b)}_{B \to A}(r) \right] \right\},$$

and the radius-specific validation score is

$$C(r) = \tanh\left(\operatorname{median}_{1 \leq b \leq B} Z_b(r) \right).$$

This criterion asks whether a metafeature learned near one spatially distributed set of fiducials predicts the anatomical target in independent fiducial territories. It assesses reproducibility of the spatial score Xw, not reproducibility of individual coefficients.

Let

$$C_{\max} = \max_r C(r).$$

We select the largest candidate radius satisfying

$$C(r) \geq C_{\max} - 0.01.$$

The tolerance favours inclusion of more tissue whenever held-out performance is practically indistinguishable from the best observed value.

Let $\widehat{r}$ denote the selected radius and define the union of all selected peri-fiducial territories by

$$\mathcal{I}_{\text{fid}} = \mathcal{I}_{\{1,\ldots,F\}}(\widehat{r}).$$

The initial metafeature direction $\hat{w}^{(0)}$ is then obtained from Equations (1) and (2) using $\mathcal{I} = \mathcal{I}_{\text{fid}}$, with sign chosen so that its training correlation with s_{T_0} is positive. The full-slide MSI pattern used for geometric alignment is

$$q^{(0)} = X\hat{w}^{(0)}.$$

2.6 Smooth thin-plate spline alignment

Starting from T_0, we estimate a globally smooth residual correction

$$T_1(u) = T_0(u) + \Delta^{\mathrm{TPS}}(u).$$

This is the principal fine-alignment step. The TPS is intentionally restricted to broad, smoothly varying deformation; local discontinuity associated with tissue tears is treated only afterwards.

Let $c_1, \ldots, c_K \in \mathbb{R}^2$ denote TPS centres in the original MSI coordinate system. To keep optimisation feasible for slides with several hundred thousand MSI spots, we use a coarse square Cartesian grid with the same centre spacing in both coordinate directions and at most

$$K_{\max} = 150$$

centres in total. Let W and H be the width and height of the bounding box of the valid MSI positions, expressed in MSI-pixel units. We choose a single integer grid spacing d_{TPS}, also in MSI-pixel units, as the smallest positive integer for which

$$\left(\left\lceil \frac{W}{d_{\mathrm{TPS}}} \right\rceil + 1\right)\left(\left\lceil \frac{H}{d_{\mathrm{TPS}}} \right\rceil + 1\right) \leq K_{\max}.$$

The numbers of columns and rows then follow as

$$K_x = \left\lceil \frac{W}{d_{\mathrm{TPS}}} \right\rceil + 1, \qquad K_y = \left\lceil \frac{H}{d_{\mathrm{TPS}}} \right\rceil + 1, \qquad K = K_x K_y \leq 150.$$

The centres are placed on this square grid, centred on the MSI bounding box; the outermost centres may therefore lie slightly beyond the bounding box. This gives a simple isotropic basis whose density is determined solely by the 150-centre cap. The regular, coarse placement is deliberate: adapting the basis to local tissue irregularities would make it easier for the TPS to absorb structures that should instead be treated as local defects.

For displacement coordinate $\ell \in \{1, 2\}$, write

$$\Delta_\ell^{\mathrm{TPS}}(u) = \sum_{k=1}^{K} \theta_{k\ell} \phi\left(\|u - c_k\|_2\right),$$

with

$$\phi(r) = r^2 \log r, \qquad \phi(0) = 0.$$

No additional affine term is included. Translation, rotation, global scaling and shear are assumed to have been handled by the fiducial-based transformation T_0; the fine-alignment TPS is reserved for non-affine residual deformation.

To remove the affine null space of the radial TPS representation, the coefficients satisfy

$$\sum_{k=1}^{K} \theta_{k\ell} = 0, \qquad \sum_{k=1}^{K} \theta_{k\ell} c_{k,1} = 0, \qquad \sum_{k=1}^{K} \theta_{k\ell} c_{k,2} = 0, \qquad \ell \in \{1, 2\}.$$

Let

$$\Theta = (\theta_{k\ell})_{k=1,\ldots,K;\, \ell=1,2} \in \mathbb{R}^{K \times 2}$$

and define the matrix $K_{\mathrm{TPS}} \in \mathbb{R}^{K \times K}$ by

$$K_{\mathrm{TPS},jk} = \phi\left(\|c_j - c_k\|_2\right).$$

Up to the conventional multiplicative constant implied by the chosen radial basis, the TPS bending energy is

$$E_{\mathrm{bend}}\left(\Delta^{\mathrm{TPS}}\right) = \mathrm{tr}\left(\Theta^{\top} K_{\mathrm{TPS}} \Theta\right)$$

[3]. It is small for slowly varying displacement fields and increases with local curvature.

For a fixed regularisation value λ_{TPS}, the TPS coefficients are estimated by

$$\widehat{\Delta}^{\mathrm{TPS}} = \arg\max_{\Delta^{\mathrm{TPS}}} \left\{ \mathrm{cor}\left(q^{(0)}, s_{T_0+\Delta^{\mathrm{TPS}}}\right) - \lambda_{\mathrm{TPS}} E_{\mathrm{bend}}\left(\Delta^{\mathrm{TPS}}\right) \right\},$$

subject to the side constraints above. The correlation term rewards cross-modal agreement, whereas the bending penalty prevents the TPS from explaining local molecular–histological disagreement by excessive curvature.

The scale of the TPS penalty is fixed by the same geometric premise used for initial metafeature learning: the fiducial neighbourhoods are the part of the slide in which T_0 is already most trustworthy. We write

$$\lambda_{\mathrm{TPS}} = 2^k$$

and fit the TPS for candidate integer values of k. For each fit, define

$$D_{\mathrm{fid}}(k) = \mathrm{median}\left\{ \left\| \widehat{\Delta}^{\mathrm{TPS},(k)}(u_i) \right\|_2 : i \in \mathcal{I}_{\mathrm{fid}} \right\},$$

where coordinates are measured in MSI-pixel units. We choose the weakest regularisation on the power-of-two grid satisfying

$$D_{\mathrm{fid}}(k) \leq 1.$$

Thus, the TPS is allowed to use as much flexibility as possible while keeping at least 95% of the residual motion in the peri-fiducial region below one MSI pixel spacing. This gives the penalty a direct geometric interpretation and avoids selecting it by maximising the same cross-modal agreement used in the registration objective.

The search starts at $k = 0$. If the criterion already holds, k is decreased in unit steps until it first fails and the preceding passing value is retained. If it fails at $k = 0$, k is increased until the criterion first holds. Each candidate optimisation is initialised at zero residual displacement.

Finally,

$$T_1(u) = T_0(u) + \widehat{\Delta}^{\mathrm{TPS}}(u)$$

denotes the selected globally smooth alignment. The TPS kernel matrix contains at most $K_{\max}^2 = 22{,}500$ entries, while evaluation at all MSI positions involves an $n \times K$ basis matrix with $K \leq 150$. For large slides, this basis matrix is evaluated in blocks during optimisation rather than stored in full.

2.7 Tear detection

The smooth TPS correction described above assumes that the residual deformation varies continuously across the tissue. Tissue preparation, however, may introduce tears or local separations for which this assumption is inappropriate. In such regions, tissue on opposite sides of a tear may have moved relative to each other, and forcing the registration to remain smooth across the discontinuity can distort the alignment on both sides. Similar problems occur more generally in histological image registration in the presence of sectioning artefacts [4, 5].

We therefore retain the smooth TPS alignment as the default registration and introduce a local tear-aware correction only where there is evidence that the smooth model is inadequate. The basic idea is to identify candidate discontinuities, cut the spatial smoothness graph along these discontinuities, and estimate any additional displacement separately on the resulting tissue regions. This follows the general principle of piecewise-smooth modelling, in which smoothness is retained away from a small set of discontinuities but is not enforced across the discontinuity itself [6, 7].

We first detect MSI spots of unusually rapid TPS deformation. Let

$$\Delta_i^{\mathrm{TPS}} = \Delta^{\mathrm{TPS}}(u_i)$$

denote the residual TPS displacement at MSI spot i. On the eight-neighbour graph of the original MSI grid, we quantify the local change in displacement between neighbouring spots by

$$g_{ij} = \frac{\left\| \Delta_i^{\mathrm{TPS}} - \Delta_j^{\mathrm{TPS}} \right\|_2}{\| u_i - u_j \|_2}, \qquad i \sim j.$$

For each spot, we retain the largest local displacement gradient,

$$G_i = \max_{j: j \sim i} g_{ij}.$$

A true tear cannot be represented as a discontinuity by the globally smooth TPS. Instead, the TPS is expected to accommodate such a discontinuity by a region of unusually rapid but still continuous deformation. We therefore use large values of G_i as evidence that the smooth model is locally under strain.

Because the overall magnitude of the TPS deformation may differ between slides, the values G_i are robustly standardised to obtain z-like scores,

$$Z_i^{\mathrm{TPS}} = \frac{G_i - \mathrm{median}(G)}{1.4826 \cdot \mathrm{median}_k \, |G_k - \mathrm{median}(G)|}.$$

Candidate high-strain spots are defined by

$$\mathcal{H}_{\mathrm{TPS}} = \left\{ i : Z_i^{\mathrm{TPS}} > z_{\mathrm{cut}} \right\}, \quad z_{\mathrm{cut}} = 4.$$

If the G_i followed a Gaussian distribution, $z_{\mathrm{cut}} = 4$ would correspond approximately to four standard deviations above the centre. The threshold is intentionally conservative because a high displacement gradient alone does not establish the presence of a physical tear.

The geometric evidence is complemented by information from the H&E image. Physical tears generally appear in the H&E image as bright or weakly stained regions. Brightness alone is nevertheless not specific: sinusoids, vessel lumina and other normal anatomical structures can produce similar image intensities. H&E brightness is therefore used only as a permissive anatomical gate, whereas the TPS displacement gradient provides the primary evidence for a tear.

Let $\mathrm{sat}(z)$ and $\mathrm{val}(z)$ denote the saturation and value channels at location z of the H&E image, scaled to $[0, 1]$. We define the whiteness score

$$W(z) = \mathrm{val}(z) \{1 - \mathrm{sat}(z)\},$$

so that large values correspond to bright, weakly saturated pixels. Hue is not used because it becomes unstable for nearly white pixels. We use segmented vessel lumina as a slide-specific reference for unstained or only weakly stained tissue regions. Importantly, only pixels belonging to the vessel lumen are used for this calibration; stained vessel walls are excluded. Let Ω_{lum} denote the resulting set of vessel-lumen pixels. The H&E whiteness threshold is defined as

$$t_{\text{HE}} = Q_{0.05}\left(\{W(z) : z \in \Omega_{\text{lum}}\}\right).$$

Thus, approximately 95% of vessel-lumen pixels are classified as sufficiently white to pass the gate. This deliberately permissive choice is appropriate because the brightness criterion is not intended to identify tears on its own; it only excludes high-strain regions whose H&E appearance is clearly incompatible with a tissue separation.

The corresponding binary mask is

$$B_{\text{HE}}(z) = \mathbf{1}\left\{W(z) \geq t_{\text{HE}}\right\}.$$

For MSI spot i, let $\mathcal{F}_i$ denote its approximate H&E footprint, defined as the disk centred at the TPS-aligned position $T_1(u_i)$ with radius ρ, the support radius of the Wendland kernel and approximately half the diagonal of one mapped MSI grid cell. We define the spot-level whiteness indicator by a maximum over its footprint,

$$B_{\text{HE},i} = \max_{z \in \mathcal{F}_i} B_{\text{HE}}(z).$$

Hence, the H&E criterion is satisfied whenever any part of the mapped MSI footprint overlaps a sufficiently bright region.

The initial tear-core candidate set is

$$\mathcal{C}_0 = \left\{ i : Z_i^{\text{TPS}} > z_{\text{cut}} \quad \text{and} \quad B_{\text{HE},i} = 1 \right\}.$$

Small gaps in this set may arise from image noise or incomplete staining. We therefore apply a one-MSI-pixel morphological closing to $\mathcal{C}_0$ and denote the resulting set by $\mathcal{C}$. Using the 3×3 square neighbourhood, this corresponds to a one-pixel dilation followed by a one-pixel erosion. The closed mask is then skeletonised to obtain a one-dimensional representation of the tear Γ. Thus, $\mathcal{C}$ provides a connected estimate of the tear-support region, whereas Γ is used subsequently to identify MSI neighbourhood edges that cross the tear.

To determine the region in which an additional correction may be required, we use a lower strain threshold $z_{\text{move}} = 2 < z_{\text{cut}}$ and define $\mathcal{H}_{\text{move}} = \{i \in \mathcal{N} : Z_i^{\text{TPS}} > z_{\text{move}}\}$. The higher threshold z_{cut}, together with the H&E gate, therefore identifies candidate tears, whereas the lower threshold allows the movable region to extend into the surrounding moderately elevated strain field.

Because thresholding a smooth strain field may create isolated pixels or narrow one-pixel connections between otherwise distinct regions, we apply a one-pixel morphological opening to $\mathcal{H}_{\text{move}}$. Using the same 3×3 square neighbourhood, this corresponds to a one-pixel erosion followed by a one-pixel dilation. The operation removes isolated high-strain pixels and narrow bridges while retaining broader regions of elevated strain. We denote the opened mask by $\mathcal{H}^{\circ}_{\text{move}}$. The movable region $\mathcal{U}$ is defined as the union of all connected components of $\mathcal{H}^{\circ}_{\text{move}}$ that touch $\mathcal{C}$, where touching means that the component either intersects $\mathcal{C}$ or contains an eight-neighbour of a spot in $\mathcal{C}$. Connectedness is also defined on the eight-neighbour MSI grid. Thus, a spot is allowed to move only if it belongs to a spatially coherent region of elevated TPS strain associated with a candidate tear. All remaining spots constitute the set $\mathcal{G}$ of trusted spots, so that

$$\mathcal{N} = \mathcal{G} \,\dot{\cup}\, \mathcal{U}.$$

Spots in $\mathcal{G}$ remain fixed at their TPS-aligned positions. Spots in $\mathcal{U}$ may receive an additional local displacement in the tear-aware refinement. The purpose of this binary partition is to separate regions for which the smooth TPS alignment is considered sufficient from regions in which the observed strain pattern provides evidence that an additional local correction may be necessary.

2.8 Tear-aware registration refinement

The purpose of the present step is to allow a local correction of the TPS alignment in those parts of $\mathcal{U}$ for which such a correction can be inferred from the surrounding trusted tissue.

The tear-support set $\mathcal{C}$ itself is excluded from local registration, because cross-modal correspondence within a physical tear is unreliable. On the remaining MSI spots we use the ordinary eight-neighbour graph, while suppressing diagonal corner connections across $\mathcal{C}$: a diagonal edge is removed if either of the two other corners of the corresponding 2×2 MSI square belongs to $\mathcal{C}$. This prevents spatial regularisation from coupling the two sides of a tear.

Among the spots in $\mathcal{U} \setminus \mathcal{C}$, we refine all connected components that are connected to at least one trusted spot in $\mathcal{G}$. Their union is denoted by $\mathcal{V} \subseteq \mathcal{U}$. Thus,

$$\mathcal{N} = \mathcal{G} \,\dot\cup\, \mathcal{V} \,\dot\cup\, (\mathcal{U} \setminus \mathcal{V}).$$

Here, $\mathcal{V}$ contains the locally refined spots, whereas $\mathcal{U} \setminus \mathcal{V}$ contains the tear-support spots and any disconnected tissue fragments for which no reliable local correction can be inferred. These retain their TPS-aligned coordinates. For an otherwise unfragmented section, typically $\mathcal{V} = \mathcal{U} \setminus \mathcal{C}$.

For each $i \in \mathcal{V}$, let

$$d_{\mathcal{G}}(i) = \min_{j \in \mathcal{G}} \|u_i - u_j\|_2, \qquad d_{\Gamma}(i) = \min_{j \in \Gamma} \|u_i - u_j\|_2$$

denote its Euclidean distances to the trusted region and the tear skeleton, respectively, in MSI-pixel units. Since both sets are represented on the regular MSI grid, these distance maps are obtained efficiently by standard Euclidean distance transforms of the corresponding binary masks. We define the mobility

$$m_i = \frac{d_{\mathcal{G}}(i)}{d_{\mathcal{G}}(i) + d_{\Gamma}(i)}, \qquad i \in \mathcal{V},$$

and set $m_i = 0$ for $i \in \mathcal{G}$. Thus, mobility is small close to trusted tissue and increases towards the tear.

Let $\Delta_i \in \mathbb{R}^2$ denote the additional displacement relative to the TPS transformation T_1, with $\Delta_i = 0$, $i \in \mathcal{G}$. For a candidate displacement field, define

$$T_{\Delta}(u_i) = T_1(u_i) + \Delta_i.$$

Let E_{loc} contain the retained eight-neighbour edges within $\mathcal{G} \cup \mathcal{V}$, excluding edges between two trusted spots. We regularise the additional displacement by

$$R(\Delta) = \frac{1}{|E_{\text{loc}}|} \sum_{(i,j) \in E_{\text{loc}}} \frac{\|\Delta_i - \Delta_j\|_2^2}{m_i + m_j}.$$

This is the usual quadratic graph-smoothing penalty with a spatially varying weight. Close to trusted tissue, where $m_i + m_j$ is small, changes in displacement are strongly penalised. Towards the tear, the penalty is progressively relaxed, allowing larger local adjustments while retaining a smooth displacement field.

The local registration uses the same fixed MSI metafeature $q^{(0)}$ and the same smooth H&E target as the preceding TPS alignment. We estimate

$$\widehat{\Delta} = \arg\max_{\Delta} \left[\mathrm{cor}\left(q_{\mathcal{V}}^{(0)}, s_{T_\Delta,\mathcal{V}} \right) - \lambda_{\mathrm{local}} R(\Delta) \right],$$

subject to $\Delta_i = 0$ for $i \in \mathcal{G}$.

The regularisation strength λ_{local} is calibrated using trusted tissue surrounding the detected tear regions. Starting from $\mathcal{U}$, we successively dilate its binary mask on the MSI grid and define

$$\mathcal{A}_{\mathrm{val}} = \mathrm{dilate}_r(\mathcal{U}) \cap \mathcal{G},$$

where r is the smallest dilation radius for which $|\mathcal{A}_{\mathrm{val}}| \geq |\mathcal{U}|$. Thus, the total dilated tear-associated area is approximately doubled, with the newly added trusted tissue serving as a validation region. If the available trusted tissue is insufficient to reach this size, all trusted spots reached before the slide boundary is encountered are used.

For calibration, the spots in $\mathcal{A}_{\mathrm{val}}$ are temporarily allowed to move while the surrounding trusted tissue remains fixed. We set their mobility to one and otherwise use the same correlation objective and graph-smoothing penalty as above. For candidate values $\lambda_{\mathrm{local}} = 2^k$, we choose the weakest regularisation for which

$$Q_{0.95} \left\{ \|\widehat{\Delta}_i\|_2 : i \in \mathcal{A}_{\mathrm{val}} \right\} \leq 0.5$$

MSI pixels. Thus, in tissue already considered reliably aligned after the TPS step, the local refinement is required to produce essentially no additional motion.

The final transformation is

$$\widehat{T}(u_i) = T_1(u_i) + \widehat{\Delta}_i, \qquad i \in \mathcal{G} \cup \mathcal{V}$$

No translation is estimated on $\mathcal{N} \setminus (\mathcal{G} \cup \mathcal{V})$. For quality control, we report the correlation before and after local refinement and the distribution of correction magnitudes $\|\widehat{\Delta}_i\|_2$ over $i \in \mathcal{V}$. We additionally report the residual local strain

$$r_{ij} = \frac{\|\widehat{\Delta}_i - \widehat{\Delta}_j\|_2}{\|u_i - u_j\|_2}, \qquad (i,j) \in E_{\mathrm{loc}},$$

summarised by its median, 95th percentile and maximum. Together, these quantities report how strongly the local refinement changes the TPS alignment and whether the inferred correction remains spatially smooth.

2.9 Identification of a hepatocyte nucleus-associated MSI metafeature

We next search for an MSI metafeature associated with hepatocyte nuclei. The aim is not to distinguish spatially extended nucleus-rich from nucleus-poor tissue regions. Such differences may reflect cell-type composition or other large-scale anatomical variation rather than molecular enrichment at nuclei. Instead, we search for an MSI feature combination that is locally enriched or depleted at MSI measurements containing hepatocyte nuclei relative to the surrounding hepatocyte background.

To make this analysis conservative with respect to image registration, we use only the trusted MSI spots $\mathcal{G}$ defined in Section 2.7. These spots require no tear-specific correction and remain at their TPS-aligned positions

$$p_i = T_1(u_i), \qquad i \in \mathcal{G}.$$

The analysis is therefore independent of the additional tear-aware refinement.

Let M, M_{hep} and M_{nuc} denote the H&E-derived binary masks of valid tissue, hepatocyte tissue and nuclei, respectively. The nuclear mask is obtained from the high-resolution nucleus segmentation and is used directly; no additional blob detection or spatial filtering is applied to it.

The H&E masks are projected to the spatial scale of an MSI measurement using the same locally normalised Wendland kernel K_ρ introduced above. For trusted MSI spot i, define the local hepatocyte fraction and the nuclear occupancy within hepatocyte tissue as

$$h_i = \frac{(M_{\mathrm{hep}} * K_\rho)(p_i)}{(M * K_\rho)(p_i)}, \qquad y_i = \frac{((M_{\mathrm{nuc}} \cdot M_{\mathrm{hep}}) * K_\rho)(p_i)}{(M_{\mathrm{hep}} * K_\rho)(p_i)}.$$

Thus, y_i approximates the fraction of the hepatocyte part of one MSI footprint occupied by segmented nuclei. The Wendland operation is used here only to represent the spatial averaging inherent in an MSI measurement; it is not intended as a background-removal step.

We restrict the analysis to trusted, hepatocyte-rich MSI spots,

$$\mathcal{I} = \{i \in \mathcal{G} : h_i \geq h_{\min}\}, \qquad h_{\min} = 0.9.$$

The restriction prevents vessel regions, immune-cell aggregates and other non-hepatocyte compartments from driving the metafeature. No additional exclusion around tears is required because only spots in the trusted region $\mathcal{G}$ enter the analysis.

The main potential confounder is broad spatial variation of the MSI features. In liver tissue, periportal–pericentral zonation and other large-scale patterns can dominate feature variance and may produce an apparent association with nuclear occupancy without representing local nuclear enrichment. We therefore remove a smooth local background from the MSI features, but deliberately leave the nuclear target y unchanged. The target is already derived from a segmented nucleus mask and contains no staining or image-intensity background that needs to be removed.

Because the eligible set $\mathcal{I}$ can be irregular, fragmented and poorly connected, the MSI background is estimated by local linear rather than local constant smoothing. For a spot-level signal g and each i in the set used for a particular fit, we fit a local plane $a + b^\top(p_j - p_i)$ to neighbouring spots j in the same set using Gaussian spatial weights

$$\kappa_{ij} = \exp\left(-\frac{\|p_j - p_i\|_2^2}{2\sigma_{\mathrm{bg}}^2}\right), \qquad \|p_j - p_i\|_2 \leq 3\sigma_{\mathrm{bg}}.$$

The fitted intercept $\hat{a}_i$ defines the local background $(Sg)_i = \hat{a}_i$. Local linear smoothing is used because, unlike a simple weighted mean, it can accommodate a local spatial gradient even when available neighbours lie predominantly on one side of the evaluation point. Fits with a rank-deficient local design matrix are omitted.

We use

$$\sigma_{\mathrm{bg}} = 4d_{\mathrm{MSI}},$$

where d_{MSI} denotes the typical spacing of neighbouring MSI measurements. This scale is substantially larger than the single-footprint scale represented by K_ρ. It therefore removes broad MSI variation while retaining variation extending over several neighbouring measurements.

Let $\mathcal{J} \subseteq \mathcal{I}$ denote the spots used for a particular model fit. Applying the local-background operator to every MSI feature gives

$$X_{\mathcal{J}}^{\mathrm{HP}} = (I - S)X_{\mathcal{J}}.$$

The nuclear target remains $y_{\mathcal{J}}$. Thus, the regression asks whether local departures of MSI features from their surrounding hepatocyte background predict the amount of hepatocyte nuclear material within the corresponding MSI footprint.

Even within hepatocyte-rich tissue, nuclear material occupies less area than non-nuclear hepatocyte material. We retain the continuous occupancy target and balance nuclear and non-nuclear tissue softly. Define

$$Y_1 = \frac{1}{|\mathcal{J}|}\sum_{i\in\mathcal{J}} y_i, \qquad Y_0 = \frac{1}{|\mathcal{J}|}\sum_{i\in\mathcal{J}}(1-y_i),$$

and

$$a_i = \frac{1}{2}\left[\frac{y_i}{Y_1} + \frac{1-y_i}{Y_0}\right].$$

The two occupancy components therefore contribute equally to the total observation weighting, while mixed MSI footprints are retained rather than assigned to an arbitrary binary class.

Spatial background removal changes both the mean and variance of the individual MSI features. For feature j, we therefore calculate the weighted training mean

$$\mu_j = \frac{\sum_{i\in\mathcal{J}} a_i X_{ij}^{\mathrm{HP}}}{\sum_{i\in\mathcal{J}} a_i}$$

and weighted training variance

$$\tau_j^2 = \frac{\sum_{i\in\mathcal{J}} a_i \left(X_{ij}^{\mathrm{HP}} - \mu_j\right)^2}{\sum_{i\in\mathcal{J}} a_i}.$$

The standardised background-corrected MSI matrix Z is then defined by

$$Z_{ij} = \frac{X_{ij}^{\mathrm{HP}} - \mu_j}{\tau_j}.$$

Thus, each background-corrected MSI feature has weighted mean zero and weighted variance one in the respective training set.

We estimate an unpenalised intercept and ridge coefficients from

$$(\hat{\alpha}, \hat{\beta}) = \arg\min_{\alpha,\beta}\left\{\sum_{i\in\mathcal{J}} a_i\left[y_i - \alpha - Z_{i\cdot}\beta\right]^2 + \lambda\|\beta\|_2^2\right\}, \qquad \lambda = 0.05,$$

where $Z_{i\cdot}$ denotes the i-th row of Z. Thus, $Z_{i\cdot}\beta$ is the linear MSI score for spot i.

If τ_j denotes the weighted standard deviation defined above, the coefficients are transformed back to the original standardised MSI feature scale as

$$\gamma_j = \frac{\hat{\beta}_j}{\tau_j}.$$

The nucleus-associated metafeature direction is

$$\hat{w}_{\mathrm{nuc}} = \frac{\gamma}{\|\gamma\|_2},$$

with sign chosen such that the corresponding background-corrected MSI pattern is positively associated with nuclear occupancy. The reported metafeature itself is evaluated on the original MSI measurements,

$$q_{\mathrm{nuc}} = X\hat{w}_{\mathrm{nuc}}.$$

Thus, spatial background removal and weighted standardisation affect only the criterion used to identify the metafeature direction; the reported metafeature remains an ordinary linear combination of the original standardised MSI features.

The spatial generalisation and localisation accuracy of this metafeature are assessed on held-out tissue regions as described in Supplementary Section 2.10. For every training–validation split, the quantities μ_j, τ_j, $\hat{\alpha}$ and $\hat{\beta}$ are estimated exclusively from the training region. In particular, the training-derived μ_j and τ_j are applied unchanged when standardising the corresponding held-out MSI features. Hence, the H&E-derived nuclear occupancies in the validation region do not influence transformation or scaling of its MSI measurements.

After validation, the final direction $\hat{w}_{\mathrm{nuc}}$ is re-estimated using all eligible spots in $\mathcal{I}$.

2.10 Spatial validation of nucleus-associated MSI localisation

We next assess whether the nucleus-associated MSI metafeature can localise hepatocyte nuclei in spatially independent tissue at approximately the scale of the MSI measurement. This validation is separate from parameter selection: the background scale σ_{bg} and ridge penalty λ are fixed in advance.

A regular 5×5 grid is overlaid on the image and the tiles are assigned alternately to two checkerboard sets A and B. Only spots in $\mathcal{I}$ are retained. Since the local background operator uses MSI measurements within $3\sigma_{\mathrm{bg}}$, spots within $3\sigma_{\mathrm{bg}}$ of a tile boundary are omitted from both sets. Consequently, calculation of a retained background-corrected MSI value does not use MSI measurements from the opposite checkerboard set.

The metafeature is fitted first on A and evaluated on B, and then in the reverse direction. To reduce dependence on the arbitrary grid origin, this procedure is repeated after shifting the checkerboard by half a tile horizontally, vertically, and in both directions, yielding eight directional train–validation evaluations.

For each held-out region, broad MSI background is removed using the same local-linear procedure described above, using only MSI measurements from that held-out region. The resulting background-corrected feature j is standardised as

$$Z_{ij}^{\mathrm{val}} = \frac{X_{ij}^{\mathrm{HP,val}} - \mu_j}{\tau_j},$$

where μ_j and τ_j are those estimated from the corresponding training region. No quantities derived from the held-out nuclear target are used in this standardisation.

The background-corrected nucleus-associated MSI score in the validation region is

$$q_i^{\mathrm{HP}} = Z_{i\cdot}^{\mathrm{val}}\hat{\beta},$$

where $\hat{\beta}$ is obtained from the corresponding training fit. The intercept is omitted because it is constant across spots and hence does not affect peak positions, correlations, or the ranking of peak heights. Nucleus localisation is performed on this background-corrected score rather than on the uncorrected reported metafeature q_{nuc}, because broad anatomical MSI variation is not informative about the position of individual nuclei.

To suppress isolated grid-scale fluctuations without substantially altering spatial resolution, q^{HP} is subjected to tight Gaussian smoothing with

$$\sigma_{\mathrm{peak}} = 0.5 d_{\mathrm{MSI}}.$$

Candidate nucleus positions are the local maxima of this smoothed score on the eight-neighbour MSI grid. Distinct local maxima are retained separately; no additional merging of nearby peaks is performed.

For each grid maximum, a sub-pixel peak position is estimated by fitting a quadratic surface to its 3×3 MSI neighbourhood. The fitted stationary point is used if it represents a local maximum and lies within the central MSI grid cell; otherwise the original MSI spot centre is retained.

A candidate peak is called as a putative nucleus only if its height exceeds a threshold learned from the corresponding training region. All distinct training-region peak heights are considered as candidate thresholds, and the threshold maximising the object-level F1 score relative to the H&E-derived hepatocyte nucleus centres in the training region is selected. This threshold is then fixed before any object-level evaluation in the held-out region.

In addition, we assess detector performance over the complete range of peak-height thresholds using precision–recall curves. These curves are used only to characterise the dependence of nucleus detection on the chosen operating threshold; they are not used to modify the fitted metafeature or select its regularisation parameters. Precision–recall curves are calculated separately for the eight held-out evaluations and can be summarised after interpolation to a common recall grid.

Predicted MSI peaks and H&E-derived hepatocyte nucleus centres in the held-out region are matched one-to-one by minimising total Euclidean distance, subject to a maximum admissible matching distance

$$d_{\text{match}} = d_{\text{MSI}}.$$

A matched pair constitutes a true-positive nucleus call. Unmatched H&E nuclei are counted as false negatives, and unmatched MSI peaks as false positives. We report sensitivity,

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}},$$

positive predictive value,

$$\text{PPV} = \frac{\text{TP}}{\text{TP} + \text{FP}},$$

and their harmonic mean, the F1 score. Specificity is not reported, because point detection provides no natural definition of the number of true-negative spatial positions; any resulting specificity would depend arbitrarily on the spatial discretisation used to define potential negative locations.

For every matched true-positive call k, let

$$e_k = \widehat{p}_k - p_k^{\text{HE}}$$

denote the localisation error vector between the MSI-derived peak position and its matched H&E nucleus centre. We report the distribution of

$$\|e_k\|_2$$

in both physical units and units of d_{MSI}, including its median and 95th percentile and the fractions of matched nuclei localised within $0.5 d_{\text{MSI}}$ and d_{MSI}. The mean two-dimensional error vector is reported separately to detect a systematic residual displacement between the two modalities.

Importantly, failure to detect an individual nucleus is not necessarily a registration error. A nucleus-associated metabolite need not exhibit the same degree of enrichment at every hepatocyte nucleus. Sensitivity therefore reflects both registration quality and the biological consistency of the molecular signal. In contrast, the localisation error among correctly detected nuclei provides the more direct measure of the spatial precision of the H&E–MSI correspondence.

As a complementary validation that does not depend on discrete peak calling, we calculate a shift-response surface between the held-out background-corrected MSI score and the H&E nuclear-occupancy pattern. Let $y_\delta(i)$ denote the nuclear occupancy obtained after translating the H&E nuclear-occupancy field by $\delta = (\delta_x, \delta_y)$ relative to the MSI coordinates. For each held-out region we calculate

$$C(\delta) = \mathrm{cor}\left(q^{\mathrm{HP}}, y_\delta\right)$$

over a square grid of shifts

$$\delta_x, \delta_y \in [-2d_{\mathrm{MSI}},\, 2d_{\mathrm{MSI}}],$$

with spacing $0.25d_{\mathrm{MSI}}$. The shifted occupancy field is evaluated by interpolation of the same Wendland-projected H&E field used to construct y_i.

If the two modalities are correctly aligned at approximately the nuclear scale, the association surface should attain its maximum close to zero shift and decline as the two modalities are displaced from one another. We therefore report the complete shift-response surface and its maximising displacement

$$\widehat{\delta} = \arg\max_{\delta} C(\delta),$$

together with

$$\|\widehat{\delta}\|_2.$$

For the repeated spatial splits, correlation surfaces are combined pointwise on the Fisher scale and transformed back to correlation values before reporting the overall surface.

The object-level localisation analysis and the shift-response surface provide complementary evidence. The former asks whether individual nuclei can be localised from MSI measurements in spatially independent tissue, whereas the latter tests whether nuclear-scale molecular correspondence is maximised at the inferred H&E–MSI alignment rather than after an additional systematic displacement.

References

[1] Hrishikesh D. Vinod. Canonical ridge and econometrics of joint production. *Journal of Econometrics*, 4(2):147–166, 1976. doi: 10.1016/0304-4076(76)90010-5.

[2] David Arthur and Sergei Vassilvitskii. k-means++: The advantages of careful seeding. In *Proceedings of the Eighteenth Annual ACM–SIAM Symposium on Discrete Algorithms*, SODA '07, pages 1027–1035, Philadelphia, PA, USA, 2007. Society for Industrial and Applied Mathematics. ISBN 978-0-89871-624-5.

[3] Fred L. Bookstein. Principal warps: Thin-plate splines and the decomposition of deformations. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 11(6): 567–585, 1989. doi: 10.1109/34.24792.

[4] Leslie Solorzano, Gabriela M. Almeida, Bárbara Mesquita, Diana Martins, Carla Oliveira, and Carolina Wählby. Whole slide image registration for the study of tumor heterogeneity. *Proceedings of SPIE*, 10956:109560I, 2019. doi: 10.1117/12.2513013.

[5] Mahsa Paknezhad, Sheng Yang Michael Loh, Yukti Choudhury, et al. Regional registration of whole slide image stacks containing major histological artifacts. *BMC Bioinformatics*, 21:558, 2020. doi: 10.1186/s12859-020-03907-6.

[6] David Mumford and Jayant Shah. Optimal approximations by piecewise smooth functions and associated variational problems. *Communications on Pure and Applied Mathematics*, 42(5):577–685, 1989. doi: 10.1002/cpa.3160420503.

[7] Luigi Ambrosio and Vincenzo Maria Tortorelli. Approximation of functionals depending on jumps by elliptic functionals via γ-convergence. *Communications on Pure and Applied Mathematics*, 43(8):999–1036, 1990. doi: 10.1002/cpa.3160430805.